SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description:	**Aluminum Copper square bar, alloy 2024**
Cat No. :	**42103**
Molecular Formula	Al:Cu:Mg:Mn; 93.5:4.4:1.5:0.6 wt%

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company .

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US:**001-201-796-7100 / **Europe:** +32 14 57 52 99
CHEMTREC Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Based on available data, the classification criteria are not met

Health hazards
Based on available data, the classification criteria are not met

Environmental hazards
Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements

Hazard Statements
EUH210 - Safety data sheet available on request

Precautionary Statements

2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Aluminum	7429-90-5	EEC No. 231-072-3	93.5	-
Copper	7440-50-8	EEC No. 231-159-6	4.4	Flam. Sol. 2 (H228) Skin Irrit. 2 (H315) Eye Irrit. 2 (H319) STOT SE 3 (H335)
Manganese	7439-96-5	EEC No. 231-105-1	1.5	Flam. Sol. 2 (H228)
Magnesium	7439-95-4	EEC No. 231-104-6	0.6	Flam. Sol. 1 (H228) Water-react. 2 (H261) Self-heat. 2 (H252)

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice If symptoms persist, call a physician.

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.
Ingestion	Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.
Inhalation	Remove to fresh air. Get medical attention immediately if symptoms occur.
Self-Protection of the First Aider	No special precautions required.

4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
approved class D extinguishers. Do not use water or foam.

Extinguishing media which must not be used for safety reasons
Water may be ineffective.

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

Hazardous Combustion Products
Metal oxides.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. No special precautions required.

6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system.

6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Pick up and transfer to properly labelled containers.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Avoid contact with skin, eyes or clothing. Avoid dust formation.

Hygiene Measures
Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep in a dry place. Keep away from acids.

Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK) (Germany) Storage Class/LGK 13

Switzerland - Storage of hazardous substances Storage class - SC 11/13
https://www.kvu.ch/de/themen/stoffe-und-produkte
https://www.kvu.ch/fr/themes/substances-et-produits
https://www.kvu.ch/it/temi/sostanze-e-prodotti

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits
List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE -** 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
Aluminum		STEL: 30 mg/m³ 15 min STEL: 12 mg/m³ 15 min TWA: 10 mg/m³ 8 hr TWA: 4 mg/m³ 8 hr	TWA / VME: 10 mg/m³ (8 heures). metal TWA / VME: 5 mg/m³ (8 heures).	TWA: 1 mg/m³ 8 uren	TWA / VLA-ED: 1 mg/m³ (8 horas)
Copper		STEL: 0.6 mg/m³ 15 min STEL: 2 mg/m³ 15 min TWA: 1 mg/m³ 8 hr TWA: 0.2 mg/m³ 8 hr	TWA / VME: 0.2 mg/m³ (8 heures). TWA / VME: 1 mg/m³ (8 heures). STEL / VLCT: 2 mg/m³.	TWA: 0.2 mg/m³ 8 uren TWA: 1 mg/m³ 8 uren	TWA / VLA-ED: 0.01 mg/m³ (8 horas)
Manganese	TWA: 0.2 mg/m³ (8h) TWA: 0.05 mg/m³ (8h)	STEL: 0.6 mg/m³ 15 min STEL: 0.15 mg/m³ 15 min TWA: 0.2 mg/m³ 8 hr TWA: 0.05 mg/m³ 8 hr	TWA / VME: 1 mg/m³ (8 heures).	TWA: 0.05 mg/m³ 8 uren	TWA / VLA-ED: 0.2 mg/m³ (8 horas) TWA / VLA-ED: 0.05 mg/m³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Aluminum		TWA: 1.25 mg/m³ (8 Stunden). AGW - exposure factor 2 TWA: 10 mg/m³ (8 Stunden). AGW - exposure factor 2 TWA: 4 mg/m³ (8	TWA: 1 mg/m³ 8 horas		

		Stunden). MAK TWA: 1.5 mg/m³ (8 Stunden). MAK			
Copper		TWA: 0.01 mg/m³ (8 Stunden). MAK Höhepunkt: 0.02 mg/m³	TWA: 0.2 mg/m³ 8 horas TWA: 1 mg/m³ 8 horas	TWA: 0.1 mg/m³ 8 uren	TWA: 0.02 mg/m³ 8 tunteina
Manganese	TWA: 0.2 mg/m³ 8 ore. Time Weighted Average	TWA: 0.2 mg/m³ (8 Stunden). AGW - exposure factor 8 TWA: 0.02 mg/m³ (8 Stunden). AGW - exposure factor 8 TWA: 0.2 mg/m³ (8 Stunden). MAK TWA: 0.02 mg/m³ (8 Stunden). MAK Höhepunkt: 1.6 mg/m³ Höhepunkt: 0.16 mg/m³	TWA: 0.2 mg/m³ 8 horas TWA: 0.05 mg/m³ 8 horas	TWA: 0.2 mg/m³ 8 uren TWA: 0.05 mg/m³ 8 uren	TWA: 0.2 mg/m³ 8 tunteina TWA: 0.02 mg/m³ 8 tunteina

Component	Austria	Denmark	Switzerland	Poland	Norway
Aluminum	MAK-KZGW: 20 mg/m³ 15 Minuten MAK-TMW: 10 mg/m³ 8 Stunden	TWA: 5 mg/m³ 8 timer TWA: 2 mg/m³ 8 timer STEL: 10 mg/m³ 15 minutter STEL: 4 mg/m³ 15 minutter	TWA: 3 mg/m³ 8 Stunden TWA: 10 mg/m³ 8 Stunden	TWA: 2.5 mg/m³ 8 godzinach TWA: 1.2 mg/m³ 8 godzinach	TWA: 5 mg/m³ 8 timer STEL: 10 mg/m³ 15 minutter. pyrotechnical;value calculated powder
Copper	MAK-KZGW: 4 mg/m³ 15 Minuten MAK-KZGW: 0.4 mg/m³ 15 Minuten MAK-TMW: 1 mg/m³ 8 Stunden MAK-TMW: 0.1 mg/m³ 8 Stunden	TWA: 1.0 mg/m³ 8 timer TWA: 0.1 mg/m³ 8 timer STEL: 2 mg/m³ 15 minutter STEL: 0.2 mg/m³ 15 minutter	STEL: 0.2 mg/m³ 15 Minuten TWA: 0.1 mg/m³ 8 Stunden	TWA: 0.2 mg/m³ 8 godzinach	TWA: 0.1 mg/m³ 8 timer TWA: 1 mg/m³ 8 timer STEL: 3 mg/m³ 15 minutter. value calculated dust STEL: 0.3 mg/m³ 15 minutter. value calculated fume
Manganese	MAK-KZGW: 1.6 mg/m³ 15 Minuten MAK-TMW: 0.2 mg/m³ 8 Stunden	TWA: 0.2 mg/m³ 8 timer TWA: 0.05 mg/m³ 8 timer STEL: 0.4 mg/m³ 15 minutter STEL: 0.1 mg/m³ 15 minutter	TWA: 0.5 mg/m³ 8 Stunden	TWA: 0.2 mg/m³ 8 godzinach TWA: 0.05 mg/m³ 8 godzinach	TWA: 0.2 mg/m³ 8 timer TWA: 0.05 mg/m³ 8 timer STEL: 0.6 mg/m³ 15 minutter. value calculated;exceptions possible, see footnote 9 inhalable fraction STEL: 0.15 mg/m³ 15 minutter. value calculated;exceptions possible, see footnote 9 respirable fraction

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Aluminum	TWA: 10.0 mg/m³ TWA: 1.5 mg/m³	TWA-GVI: 10 mg/m³ 8 satima. total dust, inhalable particles TWA-GVI: 4 mg/m³ 8 satima. respirable dust	TWA: 1 mg/m³ 8 hr. respirable fraction STEL: 3 mg/m³ 15 min		TWA: 10.0 mg/m³ 8 hodinách. dust
Copper	TWA: 0.1 mg/m³	TWA-GVI: 0.2 mg/m³ 8 satima. Cu fume TWA-GVI: 1 mg/m³ 8 satima. Cu dust STEL-KGVI: 2 mg/m³ 15 minutama. dust Cu	TWA: 0.2 mg/m³ 8 hr. Cu fume TWA: 1 mg/m³ 8 hr. Cu dusts and mists STEL: 2 mg/m³ 15 min STEL: 0.6 mg/m³ 15 min		TWA: 1 mg/m³ 8 hodinách. dust TWA: 0.1 mg/m³ 8 hodinách. fume Ceiling: 2 mg/m³ dust Ceiling: 0.2 mg/m³ fume
Manganese	TWA: 0.2 mg/m³	TWA-GVI: 0.2 mg/m³ 8 satima. total dust, inhalable particles TWA-GVI: 0.05 mg/m³ 8 satima. respirable dust	TWA: 0.2 mg/m³ 8 hr. Mn fume; inhalable fraction TWA: 0.2 mg/m³ 8 hr. inhalable fraction TWA: 0.05 mg/m³ 8 hr. respirable fraction TWA: 0.02 mg/m³ 8 hr. Mn fume; respirable fraction STEL: 0.15 mg/m³ 15	TWA: 0.2 mg/m³ TWA: 0.05 mg/m³	TWA: 0.2 mg/m³ 8 hodinách. inhalable fraction of aerosol TWA: 0.05 mg/m³ 8 hodinách. respirable fraction of aerosol Ceiling: 0.4 mg/m³ inhalable fraction of aerosol Ceiling: 0.1 mg/m³ respirable fraction of

			min STEL: 0.6 mg/m³ 15 min STEL: 3 mg/m³ 15 min		aerosol

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Aluminum	TWA: 10 mg/m³ 8 tundides. total dust TWA: 4 mg/m³ 8 tundides. respirable dust		TWA: 10 mg/m³ TWA: 5 mg/m³	TWA: 1 mg/m³ 8 órában. AK	STEL: 10 mg/m³ dust and powder TWA: 5 mg/m³ 8 klukkustundum. dust and powder
Copper	TWA: 1 mg/m³ 8 tundides. total dust TWA: 0.2 mg/m³ 8 tundides. respirable dust		STEL: 2 mg/m³ TWA: 0.2 mg/m³ TWA: 1 mg/m³	STEL: 0.2 mg/m³ 15 percekben. CK TWA: 0.1 mg/m³ 8 órában. AK TWA: 0.01 mg/m³ 8 órában. AK	TWA: 1.0 mg/m³ 8 klukkustundum. total dust and powder TWA: 0.1 mg/m³ 8 klukkustundum. Cu respirable fraction, fume Ceiling: 2 mg/m³ total dust dust and powder Ceiling: 0.2 mg/m³ Cu respirable dust, fume
Manganese	TWA: 0.2 mg/m³ 8 tundides. total dust TWA: 0.05 mg/m³ 8 tundides. respirable dust	TWA: 25 mg/m³ 8 hr STEL: 50 mg/m³ 15 min	TWA: 0.2 mg/m³ TWA: 0.05 mg/m³	TWA: 0.2 mg/m³ 8 órában. AK TWA: 0.05 mg/m³ 8 órában. AK	TWA: 0.2 mg/m³ 8 klukkustundum. total dust TWA: 0.05 mg/m³ 8 klukkustundum. respirable dust TWA: 1 mg/m³ 8 klukkustundum. Mn fume, respirable dust Ceiling: 0.4 mg/m³ total dust Ceiling: 0.1 mg/m³ respirable dust Ceiling: 2 mg/m³ fume, respirable dust

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Aluminum	TWA: 2 mg/m³	TWA: 5 mg/m³ inhalable fraction IPRD TWA: 2 mg/m³ respirable fraction IPRD TWA: 1 mg/m³ IPRD			TWA: 3 mg/m³ 8 ore TWA: 1 mg/m³ 8 ore STEL: 10 mg/m³ 15 minute STEL: 3 mg/m³ 15 minute
Copper	STEL: 1 mg/m³ TWA: 0.5 mg/m³	TWA: 1 mg/m³ inhalable fraction IPRD TWA: 0.2 mg/m³ respirable fraction IPRD			TWA: 0.5 mg/m³ 8 ore STEL: 0.2 mg/m³ 15 minute STEL: 1.5 mg/m³ 15 minute
Manganese	TWA: 0.2 mg/m³ TWA: 0.05 mg/m³	TWA: 0.2 mg/m³ inhalable fraction IPRD TWA: 0.05 mg/m³ respirable fraction IPRD	TWA: 0.2 mg/m³ 8 Stunden TWA: 0.05 mg/m³ 8 Stunden	TWA: 0.2 mg/m³ TWA: 0.5 mg/m³	TWA: 0.2 mg/m³ 8 ore TWA: 0.05 mg/m³ 8 ore

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Aluminum	TWA: 2 mg/m³ 0036 MAC: 6 mg/m³	TWA: 4 mg/m³ inhalable dust TWA: 1.5 mg/m³ respirable dust		TLV: 5 mg/m³ 8 timmar. NGV TLV: 2 mg/m³ 8 timmar. NGV	
Copper	TWA: 0.5 mg/m³ 1234 MAC: 1 mg/m³	TWA: 1 mg/m³ inhalable fraction TWA: 0.2 mg/m³ respirable fraction		TLV: 0.01 mg/m³ 8 timmar. NGV	
Manganese		TWA: 0.2 mg/m³ inhalable fraction	TWA: 0.2 mg/m³ 8 urah inhalable fraction STEL: 1.6 mg/m³ 15 minutah inhalable fraction	TLV: 0.2 mg/m³ 8 timmar. NGV TLV: 0.05 mg/m³ 8 timmar. NGV	

Biological limit values
List source(s):

Component	European Union	United Kingdom	France	Spain	Germany
Aluminum					Aluminum: 50 µg/g Creatinine urine (for long-term exposures: at the end of the shift after several shifts)

Component	Italy	Finland	Denmark	Bulgaria	Romania
Aluminum					Aluminum: 200 µg/L urine end of shift
Manganese					Manganese: 10 µg/L urine end of shift

Component	Gibraltar	Latvia	Slovak Republic	Luxembourg	Turkey
Aluminum			Aluminum: 60 µg/g creatinine urine not critical		

Monitoring methods
BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.
MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust
MDHS 91 Metals and metalloids in workplace air by X-ray fluorescence spectrometry
MDHS 99 Metals in air by ICP-AES

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)
See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Copper 7440-50-8 (4.4)		DNEL = 273mg/kg bw/day		DNEL = 137mg/kg bw/day

Predicted No Effect Concentration (PNEC)
See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Aluminum 7429-90-5 (93.5)				PNEC = 20mg/L	
Copper 7440-50-8 (4.4)	PNEC = 7.8µg/L	PNEC = 87mg/kg sediment dw		PNEC = 230µg/L	PNEC = 65mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Copper 7440-50-8 (4.4)	PNEC = 5.2µg/L	PNEC = 676mg/kg sediment dw			

8.2. Exposure controls

Engineering Measures
None under normal use conditions.

Personal protective equipment

Eye Protection Wear safety glasses with side shields (or goggles) (European standard - EN 166)

Hand Protection No special protective equipment required

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Natural rubber Nitrile rubber Neoprene PVC	See manufacturers recommendations	-	EN 374	(minimum requirement)

Skin and body protection Long sleeved clothing.

Respiratory Protection No protective equipment is needed under normal use conditions.

Large scale/emergency use Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: Particle filter

Small scale/Laboratory use Maintain adequate ventilation

Environmental exposure controls Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Solid Bar	
Appearance	Silver	
Odor	Odorless	
Odor Threshold	No data available	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	No information available	
Flammability (liquid)	Not applicable	Solid
Flammability (solid,gas)	No information available	
Explosion Limits	No data available	
Flash Point	No information available	**Method -** No information available
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
pH	No information available	
Viscosity	Not applicable	Solid
Water Solubility	Insoluble in water	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Vapor Pressure	<=1100 hPa @ 50 °C	
Density / Specific Gravity	No data available	
Bulk Density	No data available	
Vapor Density	Not applicable	Solid
Particle characteristics	No data available	

9.2. Other information

Molecular Formula Al:Cu:Mg:Mn; 93.5:4.4:1.5:0.6 wt%
Evaporation Rate Not applicable - Solid

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization No information available.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid

Incompatible products. Excess heat.

10.5. Incompatible materials

Oxidizing agent.

10.6. Hazardous decomposition products

Metal oxides.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;
Oral Based on available data, the classification criteria are not met
Dermal No data available
Inhalation No data available

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Aluminum	-	-	LC50 > 0.888 mg/L (Rat) 4 h
Copper	-	-	LC50 > 5.11 mg/L (Rat) 4 h
Manganese	LD50 = 9 g/kg (Rat)	-	LC50 > 5.14 mg/L (Rat) 4 h
Magnesium	LD50 = 230 mg/kg (Rat)	-	-

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;
Respiratory No data available
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable
Solid

Symptoms / effects,both acute and delayed No information available.

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects Contains a substance which is:. Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

Component	Freshwater Fish	Water Flea	Freshwater Algae
Copper	Onchorhynchys mykiss: LC50=0.15 mg/L 96h Cuprinus carpio: LC50=0.8 mg/L 96h	EC50: = 0.03 mg/L, 48h Static (Daphnia magna)	0.0426-0.0535 mg/L EC50 72 h 0.031-0.054 mg/L EC50 96 h
Manganese	LC50: > 3.6 mg/L, 96h semi-static (Oncorhynchus mykiss)		

12.2. Persistence and degradability

Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary

Persistence Insoluble in water, May persist.
Degradability Not relevant for inorganic substances.
Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential

Product has a high potential to bioconcentrate; May have some potential to bioaccumulate

12.4. Mobility in soil

Spillage unlikely to penetrate soil Is not likely mobile in the environment due its low water solubility.

12.5. Results of PBT and vPvB assessment

No data available for assessment.

12.6. Endocrine disrupting properties

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant	This product does not contain any known or suspected substance
Ozone Depletion Potential	This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products	Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.
Contaminated Packaging	Dispose of this container to hazardous or special waste collection point.
European Waste Catalogue (EWC)	According to the European Waste Catalog, Waste Codes are not product specific, but application specific.
Other Information	Do not flush to sewer.
Switzerland - Waste Ordinance	Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600 https://www.fedlex.admin.ch/eli/cc/2015/891/en

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO Not regulated

14.1. UN number
14.2. UN proper shipping name
14.3. Transport hazard class(es)
14.4. Packing group

ADR Not regulated

14.1. UN number
14.2. UN proper shipping name
14.3. Transport hazard class(es)
14.4. Packing group

IATA Not regulated

14.1. UN number
14.2. UN proper shipping name
14.3. Transport hazard class(es)
14.4. Packing group

14.5. Environmental hazards	No hazards identified
14.6. Special precautions for user	No special precautions required.
14.7. Maritime transport in bulk according to IMO instruments	Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories
Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia

(AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Aluminum	7429-90-5	231-072-3	-	-	X	X	KE-00881	X	-
Copper	7440-50-8	231-159-6	-	-	X	X	KE-08896	X	-
Manganese	7439-96-5	231-105-1	-	-	X	X	KE-22999	X	-
Magnesium	7439-95-4	231-104-6	-	-	X	X	KE-22673	X	-

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Aluminum	7429-90-5	X	ACTIVE	X	-	X	X	X
Copper	7440-50-8	X	ACTIVE	X	-	X	X	X
Manganese	7439-96-5	X	ACTIVE	X	-	X	X	X
Magnesium	7439-95-4	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed **KECL** - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Aluminum	7429-90-5	-	Use restricted. See item 75. (see link for restriction details)	-
Copper	7440-50-8	-	Use restricted. See item 75. (see link for restriction details)	-
Manganese	7439-96-5	-	-	-
Magnesium	7439-95-4	-	-	-

REACH links
https://echa.europa.eu/substances-restricted-under-reach

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Aluminum	7429-90-5	Not applicable	Not applicable
Copper	7440-50-8	Not applicable	Not applicable
Manganese	7439-96-5	Not applicable	Not applicable
Magnesium	7439-95-4	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .
Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification Water endangering class = 1 (self classification)

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Aluminum	nwg	
Copper	WGK2	Class III : 1 mg/m³ (Massenkonzentration)
Manganese	WGK2	Class III : 1 mg/m³ (Massenkonzentration)
Magnesium	nwg	

Component	France - INRS (Tables of occupational diseases)
Aluminum	Tableaux des maladies professionnelles (TMP) - RG 32 Tableaux des maladies professionnelles (TMP) - RG 16,RG 16bis

Swiss Regulations
Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).
Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Copper 7440-50-8 (4.4)	Prohibited and Restricted Substances		

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3
H228 - Flammable solid
H252 - Self-heating in large quantities; may catch fire
H261 - In contact with water releases flammable gases
H315 - Causes skin irritation
H319 - Causes serious eye irritation
H335 - May cause respiratory irritation

Legend

CAS - Chemical Abstracts Service
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
ENCS - Japanese Existing and New Chemical Substances
AICS - Australian Inventory of Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
DNEL - Derived No Effect Level
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
NOEC - No Observed Effect Concentration
PBT - Persistent, Bioaccumulative, Toxic
TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
OECD - Organisation for Economic Co-operation and Development
BCF - Bioconcentration factor
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
ATE - Acute Toxicity Estimate
VOC - (volatile organic compound)

Key literature references and sources for data
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards	On basis of test data
Health Hazards	Calculation method
Environmental hazards	Calculation method

Training Advice
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Prepared By	Health, Safety and Environmental Department
Revision Date	20-Feb-2024
Revision Summary	New emergency telephone response service provider.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

Disclaimer

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End of Safety Data Sheet